

Teaching Statement

My teaching philosophy is built on the philosophy that effective engineering education occurs when analytical reasoning is paired with hands-on exploration, computational thinking, and supportive guidance. Across my work as an instructor, mentor, and researcher, I have seen how students develop confidence when abstract concepts are linked to experiments, simulations, and structured problem-solving. I aim to build learning environments where students feel comfortable asking questions and where they can deepen their understanding of how engineering principles apply in real systems.

Throughout my career, I have taught in roles including Instructor, Graduate Teaching Assistant, and Mechatronics Lab Manager. In each setting, I have helped students move between conceptual understanding and physical implementation. I focus on teaching methods that make complex ideas intuitive, encourage multiple solution pathways, and promote a classroom atmosphere where students feel supported and are willing to engage with challenging material.

My strongest foundation in teaching began as an Instructor (Assistant Lecturer) in Mechanical Engineering at the Sri Lanka Institute of Information Technology (SLIIT). I taught courses such as Mechanics of Machines, Engineering Drawing, Solid Mechanics and Mechanical Design, Control Systems, and Mechatronics Systems. I also mentored a master's student on her research work and supervised several undergraduates during lab rotations and project-based volunteering assignments. These experiences helped me understand the variety of support students need when transitioning from coursework to independent technical work. Examples of my teaching materials are included in my teaching portfolio at thilinahwe.github.io/#/teaching.

Teaching these courses for the first time required me to develop all the lecture material on my own. Before the semester began, I created comprehensive notes based on previous materials, the recommended textbook, and my early TA experience. This gave me a solid outline, although maintaining pacing during lectures required continuous refinement. To address this, I rehearsed classes in advance, reviewed audio recordings of my lectures, and used these observations to improve clarity and flow. I refined slides, added CAD models and simple animations, and created dedicated tutorial sheets with practice problems. I also provided optional homework and maintained regular communication through email and office hours. These strategies helped me anticipate difficulties and strengthen my confidence as a first-time instructor.

My experience broadened further when I served as Mechatronics Lab Manager. I designed, built, and maintained undergraduate laboratories that used sensors, actuators, microcontrollers, and instrumentation systems. I supervised lab sessions, guided students through debugging circuits and code, and supported semester-end design projects. Developing laboratory modules helped me connect theory with hardware and reinforced the importance of safety, clear demonstrations, and structured instructional support for students with varied preparation levels.

During my Ph.D. at the University of Iowa, I expanded my instructional experience as a Graduate Teaching Assistant for the course *Novel Artificial Muscles and Sensors*. I designed MATLAB and Simulink tutorials that guided students through system behavior and dynamic analysis. Earlier roles at the University of Moratuwa allowed me to support laboratory courses in COMSOL Multiphysics modeling, manufacturing engineering, automation systems involving PLCs and microcontrollers, and control systems using NI-myRIO hardware. These experiences strengthened my ability to teach computational reasoning, simulation-based learning, and experimental validation

across a variety of engineering topics.

Across these roles, I have adopted a student-centered approach that uses computational and interactive tools to help students understand challenging material. I use simulation [notebooks](#), parameter-driven models, and CAD examples to explain key ideas in mechanics, dynamics, and control. I also encourage students to document their reasoning through structured worksheets and computational notebooks, which improves reproducibility and strengthens technical communication. My [teaching certificate](#) further helped me develop practical strategies such as providing immediate feedback during in-class exercises and short lab checkpoints.

I find that theoretical courses benefit greatly from targeted visual explanations. Many students learn more effectively when concepts are introduced through demonstrations, animations, or short illustrative examples. While some students adapt quickly to abstract formulations, visual explanations help a broader range of learners follow the progression of ideas and build confidence. By combining analytical approaches with clear visual reasoning, I can support students with varied backgrounds and help them engage more comfortably with challenging topics.

I incorporate modern teaching technologies and open-source tools into my courses. Interactive homework assignments, simple simulation scripts, and hands-on coding exercises enable students to test parameter values and visualize system behavior. I guide them in using industry-relevant platforms, including MATLAB, Python, Julia, SolidWorks, Fusion 360, LabVIEW, and microcontroller-based systems. These tools support conceptual understanding while helping students develop practical skills that prepare them for internships and engineering careers.

I have also faced challenges that shaped how I teach. Some students struggled to visualize dynamics and mechanism concepts even when they understood the mathematical steps. This encouraged me to expand my use of CAD-based demonstrations and small simulation examples to make system behavior easier to interpret. My teaching certificate also introduced me to techniques such as brief formative checks and concept sketches, which help students articulate their thinking and allow me to adjust instruction as needed.

Managing large laboratory groups was another challenge. Students progressed at different paces, and providing equal support required a structured approach. I introduced checkpoints, [guided worksheets](#), and small debugging tutorials that helped students work independently while still offering space for one-on-one mentoring.

Looking ahead, I plan to continue refining these practices through modular project-based assignments, structured open-lab hours, and interactive digital tools that allow students to experiment outside the classroom. My goal is to create environments where students of all preparation levels feel supported and where steady improvement is encouraged.

My teaching experience, together with my work in laboratory development, mentoring, and applied research, has shaped how I support learners at different stages of their development. I look forward to contributing these values to my future department and to supporting students as they grow into confident and capable engineers.

Teaching Portfolio: thilinahwe.github.io/#/teaching